

LITE: Loss-resilient Immersive Telepresence with Multimodal Semantics

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Abstract

Immersive telepresence has the potential to transform real-time communication through highly interactive and engaging experiences. Despite recent advances in reducing communication and computation costs, existing systems largely overlook packet loss, which can severely degrade the quality of experience (QoE). Recovering lost immersive content is considerably more challenging than in 2D video due to the complexity of dense 3D representations. Recovery must be both accurate and timely while minimizing the communication and computation overhead it incurs. To address these challenges, we present LITE, the first *loss-resilient* immersive telepresence system. LITE incorporates three key design principles: (1) leveraging semantic communication to transmit compact motion and audio semantics, which can be reconstructed into the remote user's immersive representation and voice, enabling fast semantic-level recovery and remaining robust to congestion-control-induced rate reductions under loss; (2) fusing audio and motion semantics via a lightweight multimodal model to achieve accurate, real-time recovery of motion semantics; and (3) encoding audio semantics from multiple past frames into succinct *neural redundancy* to enable robust recovery. We prototype LITE using a well-known parametric facial motion representation and extensively evaluate its performance across diverse networks. Our results demonstrate that LITE improves QoE by up to 109% compared with existing schemes, while sustaining real-time streaming at 30 frames per second and preserving high visual fidelity (structural similarity index measure above 0.9, where 1 indicates perfect similarity).

CCS Concepts

• **Networks** → **Application layer protocols**; • **Information systems** → **Multimedia streaming**; • **Computing methodologies** → **Mixed / augmented reality**.

Keywords

Immersive Telepresence, Real-time Communication, Packet Loss Recovery, Semantic Communication, Quality of Experience, Multimodal Systems



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ACM ISBN 979-8-4007-2467-1/26/08
<https://doi.org/10.1145/3789240.3829116>

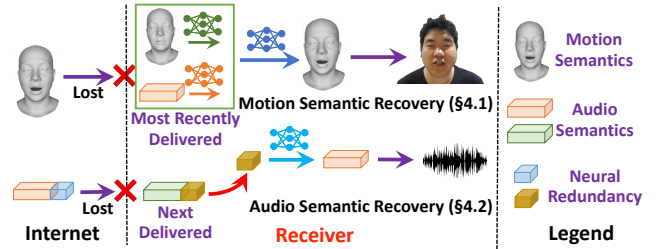


Figure 1: Key components of LITE. Lost motion and audio semantics are accurately and rapidly recovered at the receiver to reconstruct the remote user's immersive representation and voice.

ACM Reference Format:

Ruizhi Cheng, Harshvardhan Takawale, Nan Wu, Nirupam Roy, Sennur Ulukus, Matteo Varvello, Eugene Chai, and Bo Han. 2026. LITE: Loss-resilient Immersive Telepresence with Multimodal Semantics. In *ACM SIGCOMM 2026 Conference (SIGCOMM '26)*, August 17–21, 2026, Denver, CO, USA. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3789240.3829116>

1 Introduction

Immersive telepresence, a top use case in the envisioned 6G era [18], has the potential to transform real-time communication (RTC) [31, 46, 48] by delivering highly engaging and interactive user experiences. Unlike traditional 2D videos, immersive content represents 3D objects and scenes, enabling users not only to change viewing directions but also to move freely in 3D space. This unique feature has driven the wide adoption of immersive content across diverse fields, such as healthcare, education, and professional training [64]. Thus, recent years have witnessed growing initiatives to advance telepresence that delivers *live* immersive content and optimize the quality of experience (QoE) [13, 27, 38], with early commercialization efforts such as Google Beam [5].

Despite these advances, existing immersive telepresence systems largely overlook a fundamental challenge in RTC: *packet loss*. Packet loss is particularly damaging for RTC because it often manifests as visible stalls, freezes, or missing content that directly degrades QoE [8, 15, 41, 47]. This problem arises from two common sources. First, packets may be dropped during transmission, for instance, over unstable wireless and mobile networks [24]. Second, frames

may miss their playback deadlines due to the stringent end-to-end latency requirements of RTC (e.g., <100 ms one-way [13, 47]). Such deadline misses have the same impact as packet/frame losses.

Traditionally, loss resiliency is provided by retransmission and/or forward error correction (FEC) [2]. However, both are widely regarded as ineffective for interactive RTC applications [8, 15, 47, 55]. Retransmission increases delay and frequently causes frames to arrive after their deadlines [8, 47]. FEC introduces redundancy to enable recovery, but selecting an appropriate protection level is difficult [15]. More fundamentally, conventional retransmission and FEC rely on sender-side decisions that cannot dynamically adapt to real-time, mid-transit network fluctuations where losses actually occur. As a result, they struggle to react quickly and accurately to loss events that unfold at the receiver. This motivates *receiver-side* recovery mechanisms that can accurately reconstruct missing content within the playback deadline.

Recent work on 2D video conferencing explored receiver-side recovery by training deep-learning models to reconstruct missing frames [15, 41]. These approaches exploit *temporal structure* in video streams: the model learns relationships across adjacent frames and uses past observations to infer/predict missing content. Beyond loss recovery, learning-based temporal modeling has also been adopted in other parts of streaming systems, such as rate adaptation [43] and congestion control [6], indicating that receiver-side, learning-based recovery is a promising direction for immersive telepresence. However, recovering immersive 3D content is fundamentally different from 2D video and introduces several unique challenges that must be carefully addressed in system design:

- **Tight Recovery Budget.** Processing 3D content is computationally expensive (e.g., ~80 ms) [13, 38], leaving little room for recovery within the stringent latency budget for immersive telepresence. As a result, any recovery mechanism should be lightweight enough to reconstruct missing content within only a few milliseconds.
- **Heterogeneous Content Formats.** Unlike 2D video, which has a uniform representation (pixels), immersive content can be represented in multiple formats, such as raw 3D data (e.g., point clouds [27]) and high-level *semantics* [13] (§2.1). They entail different processing pipelines and incur drastically different communication and computation overhead, which directly shape the design space of loss recovery. Thus, loss-resilient immersive telepresence must systematically study these trade-offs and choose an appropriate content representation.
- **Difficulty of Modeling Temporal Redundancy.** Accurately exploiting temporal redundancy across frames is fundamental to deep-learning-based loss recovery. However, immersive 3D content is high-dimensional, and rendered frames vary dynamically with the viewer's shifting viewport [30], making it challenging to establish stable frame-to-frame correspondences.

The above challenges indicate that, to design an effective loss-resilient immersive telepresence system, we must develop a recovery model that *accurately captures temporal relationships across 3D frames while carefully balancing communication and computation costs*. To achieve this ambitious goal, we begin with a motivational study (§2.2), which uncovers the following key design insights. First, among the various paradigms, semantic communication [12], which transmits compact semantics rather than raw 3D content, is better

suited to loss-resilient telepresence. The compact nature of semantics (more than e.g., 1000× smaller than raw 3D content [13]) offers two major benefits: (i) semantics are less impacted by congestion-control-induced reductions of the sending rate under packet loss; and (ii) inference over motion semantics, which is used to reconstruct the remote user's immersive representation, is significantly faster (3 ms) than dense 3D content (>60 ms).

Second, relying solely on historical motion semantics yields poor reconstruction quality, highlighting the difficulty of modeling temporal structure from 3D data. In contrast, audio semantics, which are essential for reconstructing the sender's voice (§2.1), offer a zero-cost auxiliary signal for robust motion recovery. However, effectively capitalizing on audio semantics demands a carefully tailored model design.

Motivated by these insights, we present LITE, which, to the best of our knowledge, is the first loss-resilient immersive telepresence system (Figure 1). LITE builds on semantic communication to realize the above benefits, and further achieves accurate and rapid recovery of lost motion and audio semantics. Specifically, LITE incorporates the following innovations within a holistic system design.

Multimodal Recovery of Motion Semantics (§4.1). Recovering motion semantics involves a fundamental trade-off. Although their compactness brings several benefits for loss recovery, as shown in our motivational study, it also limits the information available for content reconstruction. Coupled with the constrained capacity of lightweight models, this makes accurate recovery particularly challenging. To address this issue, our key innovation is to *incorporate audio semantics as a reliable auxiliary signal for the recovery of motion semantics*. Specifically, we design a multimodal loss-recovery model that incorporates the most recent audio and motion semantics. To fully exploit audio semantics while preserving motion dynamics, LITE employs dedicated encoders for each modality, followed by a fusion network that effectively integrates them. This structured approach captures temporal dependencies in motion while utilizing audio cues, enabling accurate recovery of motion semantics using a lightweight model.

Audio Semantic Recovery via Neural Redundancy (§4.2). In addition to recovering motion semantics, LITE must robustly recover audio semantics to preserve speech quality and, more importantly, to provide stable auxiliary input for motion recovery. Unlike deep-learning-based approaches for predicting lost frames, audio semantics offer a unique opportunity to leverage the ground truth. This is possible because audio semantics are sampled at a high frequency (e.g., every 13.33 ms at 75 frames per second (FPS) [19]), and encoding/decoding incurs almost no delay [19, 71]. Thus, within a 100 ms one-way latency budget, it is feasible to wait for up to seven subsequent frames ($t+1$ to $t+7$) to recover a lost frame t . To leverage this insight, we design a scheme in which each transmitted frame carries information about multiple prior frames, enabling recovery of lost frames. For example, if frame t includes information about frames $t - \ell$ to $t - 1$, upon successful delivery of t , any missing frames within that range can be reconstructed. To reduce the transmission overhead introduced by carrying additional information, LITE introduces *neural redundancy* to compress audio semantics of prior frames into a succinct representation, enabling efficient and accurate recovery.

Implementation and Evaluation of LITE (§5, §6). We build a prototype of LITE and conduct a comprehensive evaluation using both controlled-loss experiments and real-world packet-loss traces [25, 47] on an immersive telepresence dataset collected through an IRB-approved user study. Additionally, we conduct separate user studies to assess QoE in terms of both visual and audio quality. For immersive content delivery, we compare LITE against nine baselines formed by combining three immersive telepresence systems, MagicStream [13], FarfetchFusion [38], and MetaStream [27], with three representative loss-recovery schemes: Tambur [55], FlexFEC [70], and ULPFEC [39]. This setup enables a comprehensive evaluation of whether LITE can consistently outperform existing loss-recovery approaches across diverse immersive streaming systems while remaining compatible with different delivery pipelines. For audio delivery, we compare LITE against three loss-recovery schemes, including both deep-learning-based and FEC-based approaches.

We summarize our key experimental results under the 100 ms per-frame playback-deadline constraint.

- On real-world traces [25, 47], LITE achieves > 0.9 structural similarity index measure (SSIM) [65], indicating a good visual quality [16]. In contrast, all baselines achieve only $SSIM < 0.86$ (poor quality). Under controlled loss rates, LITE maintains $SSIM > 0.9$ even at 30% loss rate, whereas all baselines' SSIM drops below 0.6.
- We further evaluate audio-visual synchronization using LSE-D (lip-sync error-distance) [54], a widely used lip-sync metric in which lower values indicate better synchronization. We report the LSE-D gap to the ground truth. On real-world traces, LITE's gap is at most 0.2, whereas the baselines' gaps exceed 3.5.
- LITE sustains 30 FPS streaming on average, meeting the real-time requirements of immersive telepresence [13, 38]. In contrast, baseline systems have an average FPS below 20 under real-world traces, with large variance, resulting in unsmooth playback.
- Our user study indicates that LITE yields better QoE than baselines, with up to 109% and 81% improvement in visual and audio quality, respectively.

Beyond these key results, we also demonstrate that LITE outperforms all baselines under a relaxed 150 ms per-frame playback-deadline constraint, incurs only 3 ms inference latency per frame, supports up to 10 concurrent users on a single edge device without performance degradation, and operates robustly across multiple transmission protocols and congestion control algorithms.

In this paper, we focus on recovering motion semantics (*i.e.*, FLAME parameters [40]) that will be introduced in §2.1 for the facial area, as detailed facial expressions are essential for immersive telepresence [37, 38]. Moreover, Apple's FaceTime, which supports spatial persona (arguably the first commercial mobile immersive telepresence application) on the Vision Pro headset, focuses on the facial region [14]. We note that the high-level idea of LITE can potentially be extended to recovering full-body immersive content, as discussed in §7. We also examine the deployment considerations of LITE in §7. This work does not raise any ethical issues.

2 Background and Motivation

2.1 Background

Immersive Telepresence typically utilizes multiple cameras positioned at various angles to capture diverse perspectives, which

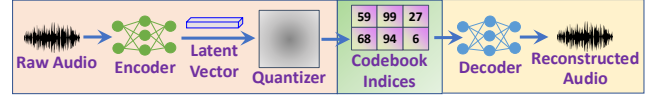


Figure 2: A paradigm of the neural audio codec.

facilitates the creation of free-view 3D models [13, 27]. In this study, we focus primarily on users as the main subjects of immersive content, as they represent the core element of telepresence systems [13, 14, 27, 52]. As a live interactive streaming system, immersive telepresence faces the challenge of packet loss, a common issue in real-world networks that can significantly degrade QoE [15, 47, 55]. Despite its impact, this issue has been largely overlooked by previous efforts [13, 27, 37, 38, 52].

Semantic Communication extracts and transmits only the essential information (*i.e.*, semantics) from the original data, rather than delivering raw content, to reduce bandwidth consumption [12]. For immersive telepresence, a prior effort, MagicStream [13], demonstrated that the human body can be effectively represented through motion and color semantics. It transmits parameters of human models, such as FLAME [40], as motion semantics. Meanwhile, lighting information is captured and transmitted as color semantics. At the receiver, MagicStream employs a neural renderer to reconstruct immersive representations of the sender from the received semantics.

Audio Semantics. Delivering high-quality audio is essential for a truly immersive experience. In this paper, we leverage emerging neural audio codecs [19, 71] to transmit audio. Their design principles follow an *encoder–quantizer–decoder* paradigm (Figure 2), aligning with the principles of semantic communication.

Specifically, these methods incorporate:

- **Encoder:** A deep neural network at the sender that converts each raw audio segment x into a feature vector $z \in \mathbb{R}^d$.
- **Residual Vector Quantization (RVQ) [9]:** A quantizer $Q(\cdot)$ that maps z into discrete codebook indices $z_q = Q(z) \in \{\Omega\}^N$, where each Ω is a pre-trained, fixed-size codebook shared with the encoder and decoder.
- **Decoder:** A deep neural network at the receiver that converts these indices back into feature vectors (using the codebook) and synthesizes the final audio waveform $\hat{x}$.

In this setup, only the compact codebook indices z_q are transmitted, which we can treat as *audio semantics*.

2.2 Motivation

As presented in §1, deep-learning-based loss recovery offers a promising approach for immersive telepresence. In this section, we examine two central design questions: (1) What transmission paradigm is most suitable for loss-resilient immersive telepresence? and (2) how can loss-recovery accuracy be improved without introducing excessive communication and computation overhead? We use the dataset and network setup introduced in §6, and conduct experiments using packet-loss traces released by Hairpin [47].

Content Type for Transmission and Recovery. Immersive telepresence typically transmits either raw 3D content, such as RGB-D or point clouds, or high-level semantics. To compare these approaches, we evaluate three representative systems: FarfetchFusion (FF) [38] for RGB-D, MetaStream (ME) [27] for point clouds, and MagicStream (MA) [13] for semantics. In the following, we investigate

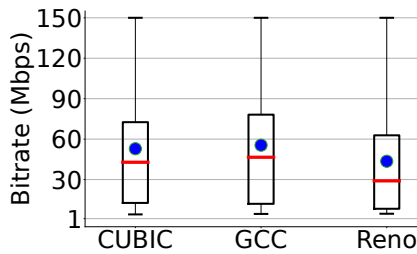


Figure 3: Sending bitrates of CUBIC [29], GCC [11], and Reno [23] over a 150 Mbps link under the packet loss trace from Hairpin [47].

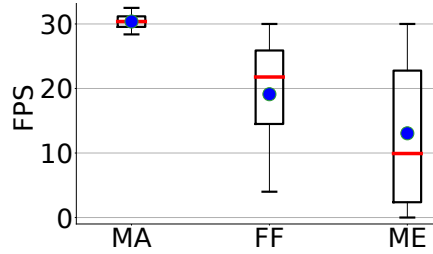


Figure 4: FPS of MagicStream (MA) [13], FarfetchFusion (FF) [38], and MetaStream (ME) [27] when operating under the same packet loss trace from Figure 3.

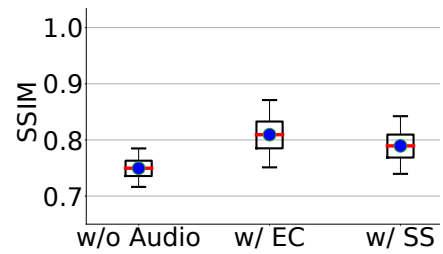


Figure 5: SSIM of recovering motion semantics using (i) previous ones without audio, (ii) with EnCodec (EC) [19], and (iii) with SoundStream (SS) [71].

their performance from two perspectives: (1) interaction with congestion control during packet loss, and (2) computation delay in deep-learning-based recovery.

(1) *Impact of Congestion Control during Packet Loss.* Congestion control is an important component of communication systems, aiming to regulate the sending rate to prevent network overload. Despite design differences, existing algorithms [11, 23, 29] share two key principles: (i) they interpret packet loss as a signal of congestion and reduce the sending rate accordingly, and (ii) they operate at the sender side. Thus, generally speaking, *under the same link capacity, systems with lower throughput demands are inherently more resilient to loss*, since they have less chance of being affected when congestion control reduces the sending rate.

To verify this in the context of immersive telepresence, we evaluate three representative congestion control algorithms: GCC [11], which is commonly used in live video systems [46, 47], as well as CUBIC [29] and Reno [23] that are widely deployed in TCP/QUIC [33, 50]. We configure the target sending bitrate (link capacity) as 150 Mbps, which is sufficient to support all systems under normal operation. Figure 3 shows the sending bitrate of three algorithms under loss. Each boxplot reports the 5th, 25th, 50th, 75th, and 95th percentiles, with blue dots indicating the mean. We observe that all three algorithms significantly reduce their sending rates during packet loss, with average throughput below 60 Mbps and a large variance ranging from ~ 1 Mbps to the full 150 Mbps capacity. This oscillation occurs because congestion control ramps up the bitrate between loss events, and then sharply reduces it when loss recurs.

Such unstable, low-bitrate connections severely affect systems that transmit raw 3D content. Figure 4 shows FPS results using GCC. Real-time telepresence requires 30 FPS [13, 38], yet both FarfetchFusion and MetaStream exhibit substantial drops in frame rate, with averages below 20 FPS and the 5th percentile below 5 FPS. In contrast, MagicStream remains unaffected since its operating bitrate is under 1 Mbps. This demonstrates that semantic-based transmission is inherently more robust under congestion control. We note, however, that semantics are still subject to packet loss during transmission, which requires effective receiver-side recovery.

(2) *Processing Delay in Deep-learning-based Recovery.* Next, we examine the computation overhead of deep-learning-based recovery for different content types. We use the recovery models described in §5 and Appendix A to recover lost frames from previous ones. Experiments are conducted on a machine equipped with an NVIDIA RTX 4090 GPU. We observe that recovering dense raw 3D content

is prohibitively slow. For example, processing a point cloud requires over 60 ms per frame. In contrast, semantics can be processed in 3 ms per frame due to their compact size.

In summary, semantic-based transmission is fundamentally more suitable for loss-resilient telepresence systems, since it is less affected by bitrate fluctuations under congestion control and its compact representation enables timely recovery within playback deadlines.

Visual Quality. Beyond timely recovery, we must ensure that recovered content preserves visual fidelity. To assess this, we compute SSIM [65] between rendered results from recovered semantics (using MagicStream’s rendering pipeline [13]) and ground-truth images captured at the sender. We design a model that recovers lost motion semantics solely from previous frames. The window size, meaning the number of previous frames utilized for loss recovery, is configured as 3, which yields the best results. As shown in Figure 5, this approach achieves an average SSIM of only 0.75. *This analysis reveals that motion-only recovery is inadequate, highlighting the need for an auxiliary modality to further enhance visual quality. However, such approaches cannot come at the expense of excessive communication and computation overhead.*

Using Audio Semantics to Aid Motion Semantic Recovery.

Next, we explore using audio semantics as a complementary modality for motion recovery. This is motivated by prior studies that demonstrate acoustic cues are highly correlated with facial expressions [69] and even full-body motion [51]. Moreover, given that audio semantics are already transmitted for speech and are compact, using them for motion recovery adds negligible communication and computation overhead.

Inspired by this, we design a multimodal model, which concatenates features from prior audio semantics (window size = 7, the configuration yielding the best results) with motion semantics, to jointly recover lost motion semantics. We evaluate two audio semantic choices, EnCodec [19] and SoundStream [71]. As shown in Figure 5, incorporating audio semantics can improve recovery quality (e.g., > 0.8 SSIM on average with EnCodec). Moreover, because audio semantics are compact, this improvement results in negligible computational overhead, allowing recovery to still meet real-time deadlines. These findings reveal that audio semantics provide a valuable auxiliary modality for motion-semantic recovery.

However, two key challenges remain. First, although the multimodal recovery model improves visual quality, its SSIM remains below 0.9, the threshold for good quality [16]. Thus, further improving the quality of recovery is essential. Second, audio streams may

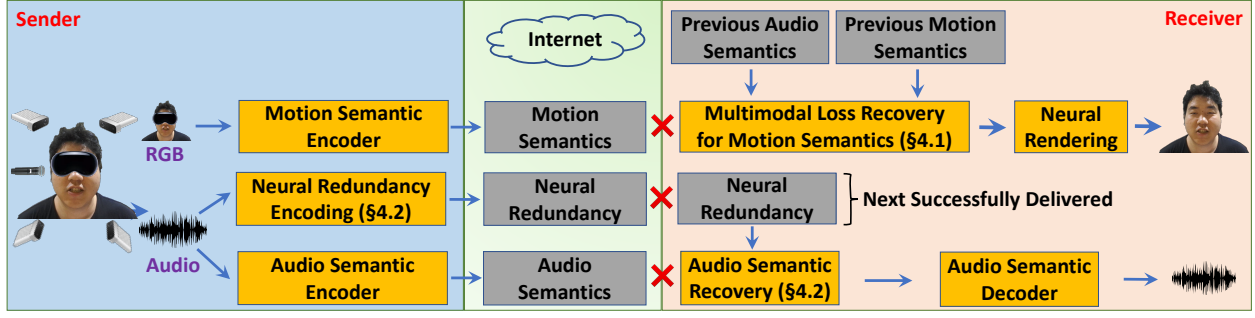


Figure 6: System architecture and workflow of LITE. We depict only one-way communication for simplicity. The reverse direction follows the same structure as shown in this figure.

also be lost during transmission. Hence, ensuring robust recovery of audio semantics is necessary.

These challenges motivate LITE’s design: (1) strategically modeling cross-modal features so that audio semantics can effectively support motion recovery, and (2) integrating a robust recovery mechanism to protect audio semantics.

3 Overview of LITE

Figure 6 shows the architecture of LITE, a loss-resilient immersive telepresence system. At the sender, LITE extracts facial motion semantics from RGB data and audio semantics from audio waveforms. In parallel, it employs a redundancy encoder to generate succinct neural redundancy from prior audio semantics and transmits it alongside the current frame’s audio semantics. At the receiver, if motion semantics are lost, LITE employs a multimodal recovery model that integrates the most recent audio semantics with motion semantics to recover the missing content (§4.1). Then, LITE uses a neural rendering model [13] to render the remote user’s immersive 3D representation from the recovered motion semantics. If audio semantics are lost, LITE reconstructs them using the neural redundancy embedded in a subsequently delivered frame (§4.2). The recovered audio semantics are then decoded into audio waveforms.

Before telepresence begins, LITE conducts a brief user profiling phase lasting 30 seconds to gather data for offline training/fine-tuning, which is commonly required in immersive telepresence systems [13, 38]. The collected data is used to train/fine-tune the recovery model for motion semantics (§4.1), the encoder/decoder of neural redundancy for audio semantics (§4.2), and the neural rendering model. Training the loss-recovery models in LITE is lightweight. It takes < 2 minutes for a new user and can be further reduced for returning users by fine-tuning based on the existing profile (§6.5).

Due to resource limitations of mobile headsets, LITE offloads computationally intensive tasks, such as loss recovery and neural rendering of motion semantics, to an edge device such as a PC with a commodity GPU. Such a design is widely used in immersive content delivery [13, 66, 72].

4 System Design of LITE

4.1 Multimodal Recovery of Motion Semantics

Insights and Challenges. Our motivational study (§2.2) demonstrates that audio semantics provide a valuable auxiliary modality for motion semantic recovery. However, this study also shows that

simply concatenating audio and motion semantics as inputs and then directly predicting the lost motion semantics is insufficient.

Our further investigations reveal that two main factors lead to this inefficiency. First, while audio waveforms correlate strongly with facial expressions [42], LITE operates on *high-level semantic* representations of audio and motion. These two modalities differ vastly in scale, dimensionality, and information density, rendering naive concatenation ineffective at extracting the subtle acoustic cues required for robust motion recovery. Second, audio semantics are generated at a higher frame rate (75 FPS) compared to motion semantics (30 FPS). Direct concatenation without proper handling risks overemphasizing audio features, thereby overshadowing the sparser motion semantics. The challenge, therefore, is how to effectively combine audio and motion semantics in a balanced manner that preserves the unique contributions of each modality while leveraging their cross-domain correlations to aid the recovery of motion semantics.

Solution. To address this challenge, we design a multimodal recovery model with two dedicated encoders, one for audio semantics and another for motion semantics, followed by a fusion network that combines their outputs to recover lost motion semantics. This design models each modality independently, preventing high-frequency audio semantics from dominating motion features, while the fusion network captures cross-modal correlations.

Following the design principle of MagicStream [13], we adopt the FLAME parameters [40] to represent motion semantics, denoted as θ_t at frame t . The historical motion window is defined as $\Theta_t^{(w_f)} = [\theta_{t-w_f}, \dots, \theta_{t-1}]$, while the corresponding audio window is $A_t^{(w_a)} = [z_{q,t-w_a}, \dots, z_{q,t}]$, where $z_{q,t}$ denotes the audio semantics (§2.1).

The two modality-specific encoders map their respective sequences into embeddings, $f_{\text{motion}} : \mathbb{R}^{w_f \times D_\theta} \rightarrow \mathbb{R}^{h_\theta}$ and $f_{\text{audio}} : \mathbb{R}^{w_a \times D_a} \rightarrow \mathbb{R}^{h_a}$. Their outputs are concatenated to form $H_t = [f_{\text{motion}}(\Theta_t^{(w_f)}); f_{\text{audio}}(A_t^{(w_a)})]$, which is then passed to the fusion network to recover the lost motion semantics, $\hat{\theta}_t = f_{\text{fusion}}(H_t)$.

This design offers three key advantages. First, isolating motion and audio semantics into separate encoders ensures balanced modeling across modalities and prevents the over-representation of audio features. Second, the fusion network learns a cross-domain representation that exploits correlations between audio and motion semantics to guide recovery. Third, missing motion frames do not affect the audio encoder. As a result, the fusion network can

still leverage the audio stream as a reliable auxiliary signal during bursty losses, supported by our loss-recovery mechanism for audio semantics (§4.2).

During training, we simulate bursty loss by randomly masking consecutive motion semantics and optimizing recovery. This masking-based training procedure is also used in the loss recovery model for 2D video streaming [15]. The objective minimizes L1 and L2 losses between the ground-truth and reconstructed FLAME parameters, as well as the vertex-to-vertex (V2V) distance between the ground-truth FLAME mesh (obtained from FLAME parameters) and the reconstructed one. Our model design is compatible with various deep learning architectures, such as LSTM [26] and Transformer [63], as presented in §6.6.

In this work, we do not focus on recovering color semantics (e.g., the lighting characteristics in MagicStream [13]), as these features remain largely stable across consecutive frames and do not demand instantaneous recovery. For example, in controlled lighting environments such as Project Starline [37] (now Google Beam [5]), lighting conditions remain consistent over time. We discuss strategies for handling the loss of color semantics in Appendix B.

4.2 Audio Semantic Recovery via Neural Redundancy

Insights and Challenges. Deep-learning-based recovery of video frames reconstructs missing data by exploiting only past information [15, 41]. Such methods are inherently *predictive*: when a frame is lost, the model attempts to infer its content relying solely on historical data.

Our key insight is that audio semantics enable a fundamentally different recovery strategy. Rather than prediction, we can recover a lost frame using ground-truth information that becomes available shortly afterward. To enable this, subsequently delivered frames should carry the necessary information to reconstruct earlier lost frames before their playback deadlines. For motion semantics, this approach is infeasible; the intensive 3D processing required at both the sender and receiver already exhausts the bulk of the end-to-end latency budget [13], leaving no margin to wait for future frames.

In contrast, audio semantics are sampled at a high frequency (e.g., 13.33 ms per frame at 75 FPS [19]), and encoding/decoding incurs almost no delay [19, 71]. This allows the system to defer recovery for several subsequent frames while still meeting the strict playback deadline. For instance, under a 100 ms deadline, a lost audio frame t can still be recovered from later frames up to $t + 7$ ($7 \times 13.3 \text{ ms} < 100 \text{ ms}$).

To exploit this property, our design lets each transmitted frame k carry additional information about prior ℓ frames: $k - \ell$ to $k - 1$. If frame k is successfully delivered, the recovery data it carries can be used to reconstruct any missing frames within that range. Thus, each frame has up to ℓ recovery opportunities, which greatly mitigates bursty losses in today's Internet [55]. The main challenge, however, is bandwidth overhead. A naïve approach would directly attach the raw semantics of ℓ prior frames to each new frame, inflating bandwidth consumption when no losses occur.

Solution. To mitigate bandwidth overhead while preserving recovery capability, we propose a lightweight *recovery redundancy encoder* R . Instead of attaching raw semantics from prior frames,

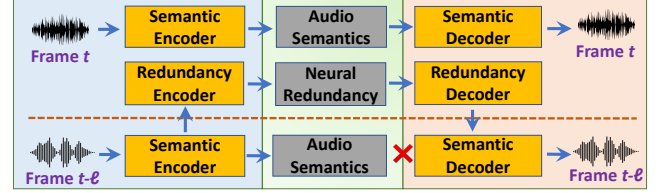


Figure 7: A paradigm of audio semantic recovery. For each audio frame t , the semantic encoder generates its audio semantics, while the redundancy encoder produces neural redundancy that summarizes semantics from prior frames ($t - \ell, \dots, t - 1$). Both the audio semantics and neural redundancy are transmitted over the Internet. If frame $t - \ell$ is lost during transmission, the redundancy decoder can reconstruct the audio semantics as long as at least one later frame from $t - \ell + 1$ to t is received.

which would quickly inflate transmission size, the encoder R compresses the semantics of prior frames into compact recovery redundancy. The key idea is to jointly exploit *temporal redundancy* across consecutive frames and *statistical redundancy* within each latent space, thereby compacting multiple audio frames into a single, low-bitrate code.

This design can be viewed as a form of neural redundancy: instead of duplicating raw semantics across frames, LITE encodes past information into compact recovery codes. Unlike FEC, whose effectiveness is tightly bound to the loss rate, recovery codes capture semantic correlations across frames and provide protection without requiring adaptation to network conditions.

Figure 7 illustrates the design. For each audio frame t , the neural audio codec encodes the waveform into a latent vector and quantizes it with RVQ into semantic indices $\mathbf{z}_{q,t}$, while the redundancy encoder, implemented as an LSTM followed by RVQ for compression, generates $\mathbf{r}_{q,t} = R(\{\mathbf{z}_{q,t-i}\}_{i=1}^{\ell})$ to summarize the prior ℓ frames. At the receiver, if frame t' is lost but a later frame $t' + i$ arrives within the redundancy window ℓ , the decoder reconstructs $\tilde{\mathbf{z}}_{q,t'} = R^{-1}(\mathbf{r}_{q,t'+i})$ and passes it to the audio semantic decoder $G(\cdot)$ to synthesize the waveform.

Beyond its low bandwidth overhead, another key advantage of our framework is its *codec-agnostic* nature. This is because the redundancy encoder operates on the quantized semantic indices produced by any neural audio codec, such as EnCodec [19] and SoundStream [71]. As a result, the same recovery mechanism can be integrated with different codecs by retraining under simulated frame losses. Specifically, during training, we sample a random subset of frames $\mathcal{M} \subset \{1, 2, \dots, T\}$, mask their latent vectors $\mathbf{z}_{q,t'}$, and rely on the neural redundancy from subsequent frames to reconstruct them. The objective minimizes a weighted combination of L1 loss, STFT loss [57], and adversarial loss [20], following standard practices for training neural audio codecs [19, 71].

5 Implementation

We use the Microsoft Azure Kinect DK cameras for data capture (see §6.1 for more details). The edge device is equipped with an NVIDIA RTX 4090 GPU unless otherwise specified. For the client device, we utilize the Microsoft HoloLens 2 headset. We use DECA [22] to extract motion semantics and color semantics. For audio processing, we employ EnCodec [19], which outperforms alternatives such as

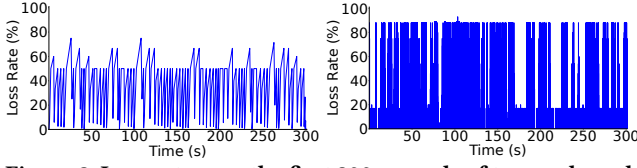


Figure 8: Loss rate over the first 300 seconds of two real-world traces: a cellular network trace from Akamai [25] (left) and a cloud-gaming trace from Hairpin [47] (right).

SoundStream [71] in our evaluation. The encoders and the fusion network used for motion semantic recovery are based on two-layer LSTMs [26]. We evaluate multiple backbone configurations in §6.6 and report the details of the selected model in Appendix A. We leverage MagicStream’s pipeline for rendering [13].

To demonstrate the effectiveness of LITE across transmission paradigms and congestion control algorithms, we implement and evaluate it over both WebRTC [7] and QUIC [36]. For WebRTC, we use GCC [10] as the congestion control algorithm, consistent with WebRTC defaults. For QUIC, we adopt the Media over QUIC (MoQ) [28] paradigm, which is specifically designed to support RTC applications over QUIC. For congestion control, we evaluate CUBIC [29] and Reno [23], both deployed in QUIC [49, 50].

6 Performance Evaluation

6.1 Experimental Setup

Datasets. Due to the lack of publicly available telepresence datasets with recorded audio, we conduct an IRB-approved user study to collect a comprehensive dataset. Our cohort consists of 6 females and 10 males, with an average age of 25.1 ± 2.9 . Participants were paired in groups of two and engaged in two 10-minute teleconferencing sessions, which represent the primary use case for telepresence systems [13, 37]. We used three Microsoft Azure Kinect DK cameras [1] per side to capture RGB data from the facial area and record audio via their integrated microphones. Using three cameras for capture is a common practice in existing immersive telepresence systems [13, 27, 38]. For each user, we use the first 80% of captured data for training and reserve the remaining 20% for evaluation.

Network Setup. We connect the sender’s and receiver’s mobile headsets to separate networks via distinct WiFi routers. We connect the edge devices to their respective routers via Ethernet. Communication between the two WiFi routers occurs over Ethernet, with an average throughput of ~ 150 Mbps. To simulate realistic Internet delays [17, 68], we introduce a round-trip latency of 40 ms between the two WiFi routers using Linux `tc` [4].

We evaluate LITE using both real-world packet-loss traces and controlled loss rates. Specifically, we employ loss traces from Akamai [25] (avg: 29.7%, std: 22.6%, max: 75%) and Hairpin [47] (avg: 10.1%, std: 21.1%, max: 93%). The Akamai traces were collected from its CDN cluster hosted in a major U.S. cellular network provider’s data center. The Hairpin traces were collected from a commercial cloud-gaming platform. We use the sample traces released by the authors of Hairpin [45]. Figure 8 shows the first 300 seconds of two real-world packet loss traces. The Akamai trace exhibits sustained periods of relatively high loss (e.g., $\sim 50\%$), resulting in a higher average loss rate of 29.7%. In contrast, the Hairpin trace has a lower average loss rate (10.1%) but displays extreme burstiness,

with instantaneous losses exceeding 90%. Overall, both traces exhibit highly bursty loss patterns, which are prevalent in today’s Internet [55], posing significant challenges for loss recovery in immersive telepresence.

Baselines. Since there is no prior work on loss-resilient immersive telepresence, we configure baselines to evaluate LITE as a combination of immersive telepresence and loss-recovery mechanisms. For immersive telepresence, we compare LITE against three state-of-the-art systems: MagicStream (MA) [13], FarfetchFusion (FF) [38], and MetaStream (ME) [27]. For loss recovery, we consider three FEC-based schemes: Tambur [55], FlexFEC [70], and ULPFEC [39], with the latter two being used in the WebRTC stack [60]. Combining these three loss-recovery schemes with the three telepresence systems yields nine baseline systems in total.

We configure all systems with the 100 ms per-frame playback deadline required by RTC systems [8, 13, 47]. One important consideration is that Tambur uses a streaming-code design that spreads parity across future frames over a decoding delay τ . For a frame i , if the receiver can wait until frame $i + \tau$ without missing the playback deadline, Tambur distributes the parity for frame i across frames i to $i + \tau$. However, as highlighted in §1, immersive telepresence leaves limited time for such delayed recovery. State-of-the-art systems already spend around 80 ms on 3D processing at the sender and receiver [13, 38], while immersive content delivery targets 30 FPS streaming (i.e., ~ 33 ms per frame) [66, 72]. Thus, under the 100 ms per-frame playback deadline, waiting for even one additional frame would generally exceed the remaining latency budget. Therefore, we configure Tambur with $\tau = 0$ for the 100 ms experiments, meaning that its parity must be decoded without waiting for future frames. To further demonstrate that LITE outperforms the baseline systems under different latency budgets, we also compare LITE under its default 100 ms deadline against all baseline systems under a relaxed 150 ms per-frame playback deadline in §6.2. This relaxed setting gives Tambur and the two WebRTC FEC schemes more time for recovery.

We do not consider retransmission, as it is widely regarded as infeasible for RTC due to strict latency constraints [8, 15]. Our experiments confirm this: in existing immersive telepresence systems [13, 27, 38], more than 95% of retransmitted frames miss the 100 ms deadline. We also exclude Hairpin [47] and Tooth [8], as both are essentially reduced to the standard FEC scheme in our problem scenarios. The detailed analysis and explanation are provided in Appendix C.

For audio content delivery, we compare LITE against three baselines: (1) LITE-L, which employs an LSTM to predict lost audio semantics, (2) Opus [61], the default WebRTC audio codec, configured with low-bitrate redundancy (LBRR), its built-in FEC scheme, and (3) Valin *et al.* [62], which employs a dedicated deep-learning model to recover lost audio streams. We employ Opus with LBRR because WebRTC recommends LBRR over FlexFEC and ULPFEC for protecting audio streams [60].

User Study. To assess user experience across LITE and the baselines, we conduct a user study with 21 participants (8 females and 13 males; average age: 29.1). We evaluate visual and audio quality independently by disabling audio during visual-quality evaluation and disabling video during audio-quality evaluation. This setup

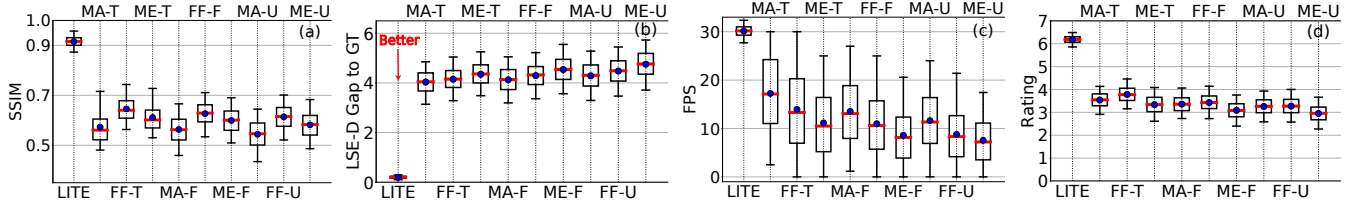


Figure 9: (a) SSIM, (b) LSE-D (lip-sync error-distance) gap to ground truth (lower is better), (c) FPS, and (d) user ratings on visual quality across LITE and nine baseline systems, formed by combining each of the three immersive telepresence systems, MagicStream (MA) [13], FarfetchFusion (FF) [38], and MetaStream (ME) [27], with each of the three loss-recovery schemes, Tambur (T) [55], FlexFEC (F) [70], and ULPFEC (U) [39], under packet-loss traces from Akamai [25]. Unless otherwise stated, all results in this and later figures are evaluated under the 100 ms per-frame playback-deadline constraint.

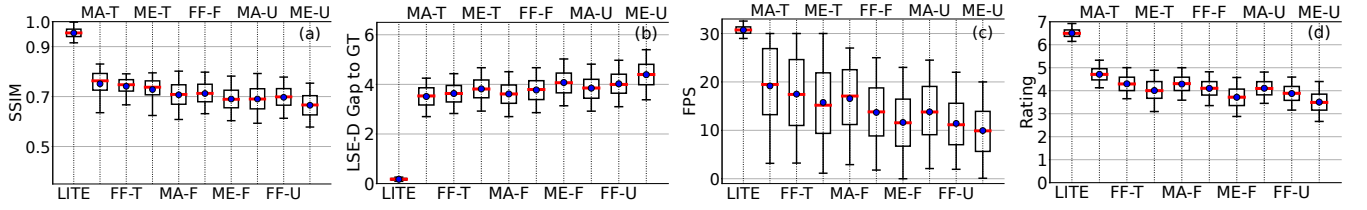


Figure 10: (a) SSIM, (b) LSE-D gap to ground truth, (c) FPS, and (d) user ratings on visual quality across LITE and the nine baseline systems under packet-loss traces from Hairpin [47].

isolates participants' perception of each modality and avoids cross-modal interference when scoring quality. Note that, in the visual-quality evaluation, the visual stream is still rendered from motion semantics recovered by the audio-conditioned multimodal model. Thus, the study still reflects the effectiveness of LITE's multimodal motion-recovery design. For immersive content, participants wear the headset to freely explore the scenes of pre-recorded telepresence sessions rendered by evaluated systems. For audio, users listen to reconstructed audio from various systems through the headset. Users rate their experience using a 7-point Likert scale (1: very bad to 7: very good) [56], followed by an exit interview.

Metrics. We evaluate LITE using the following metrics:

Visual Quality. We assess perceptual rendering quality using SSIM by comparing screenshots from the headset at the receiver with ground-truth images captured at the sender. We also evaluate audio-visual synchronization using LSE-D [54], a widely used lip-sync metric where lower values indicate better synchronization. We report the LSE-D gap relative to the ground truth, defined as the difference between the system-measured and ground-truth LSE-D values. When a video stall occurs, the receiver freezes on the last rendered frame while the ground-truth stream continues to advance. In such cases, SSIM and LSE-D are computed between the frozen frame and the corresponding new ground-truth frames.

Video Smoothness. We measure playback smoothness in terms of FPS. Telepresence requires at least 30 FPS to ensure a fluid viewing experience [13, 27, 38].

Bandwidth Consumption. We measure the total bandwidth required by each system, which is jointly affected by the transmitted content size, congestion control behavior, and FEC-induced redundancy (if applied).

User Experience. We collect user ratings to assess audio quality, a strategy widely used in prior studies [19, 71], and user-perceived visual quality in telepresence.

Latency, Scalability, and Resource Utilization. We evaluate inference latency under different edge devices, scalability with increasing numbers of concurrent users, and resource utilization on both the edge and the client.

6.2 End-to-end Evaluation

We begin by comparing LITE against various baselines. Since WebRTC is the most common transmission paradigm for live interactive streaming [44, 46], most of our evaluations are reported under WebRTC with GCC. We provide additional results for QUIC with CUBIC and Reno in §6.7. Unless otherwise stated, all results are evaluated under the 100 ms per-frame playback deadline. Figures 9, 10, and 11 present the results of LITE and the baseline systems under the Akamai trace, the Hairpin trace, and controlled loss rates, respectively. Each boxplot reports the 5th, 25th, 50th, 75th, and 95th percentiles, with blue dots indicating the mean.

Visual Quality. We first quantitatively compare visual quality using SSIM and LSE-D. As shown in Figures 9(a) and 10(a), LITE achieves an average SSIM above 0.90 across both Akamai and Hairpin traces, indicating good visual quality [16]. In contrast, the baseline systems achieve average SSIM values below 0.75, reflecting poor visual quality [16]. Moreover, LITE's LSE-D gap to ground truth is at most 0.2 under the two real-world traces, while the baselines' gaps are more than 3.5, as shown in Figures 9(b) and 10(b). Figure 11(a) further demonstrates that LITE maintains SSIM above 0.9 even under 30% packet loss, whereas all baselines' SSIM drops below 0.6 at the same loss rate. Also, LITE maintains a small LSE-D gap to ground truth even under high loss rates, remaining below 2 at 80% packet loss, as shown in Figure 11(b).

The reasons for SSIM and LSE-D degradation differ across systems. For MagicStream, drops occur when the added redundancy is insufficient to recover lost data. In such cases, the missing semantics cannot be reconstructed, leading to playback stalls. For FarfetchFusion and MetaStream, failed recovery due to insufficient

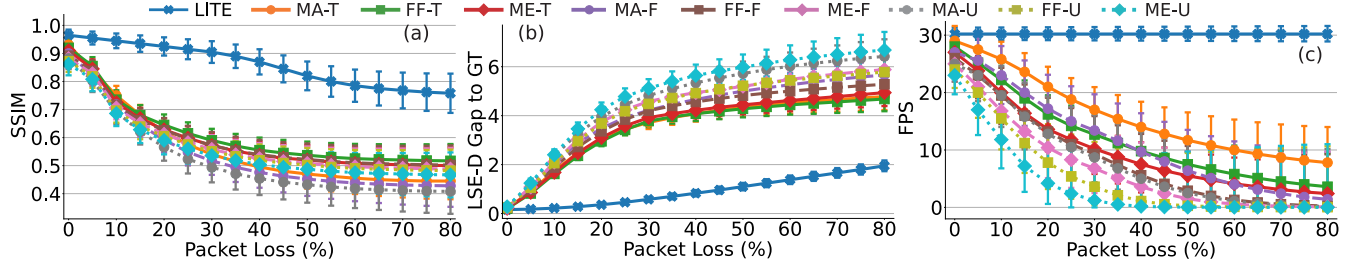


Figure 11: (a) SSIM, (b) LSE-D gap to ground truth, and (c) FPS across LITE and the nine baseline systems under different packet loss rates.



Figure 12: Qualitative comparison of LITE, FarfetchFusion with Tambur, and MetaStream with Tambur under 30% packet loss rate.

redundancy results in partial content loss. Although rendering continues, the incomplete data produces visible artifacts and degrades visual quality. Figure 12 illustrates this effect: under 30% packet loss rate, LITE can render photorealistic content, while FarfetchFusion and MetaStream (both using Tambur) exhibit noticeable artifacts (e.g., holes) in the reconstructed face.

We also observe that, for the same telepresence system, Tambur generally outperforms FlexFEC and ULPFEC. This trend holds across all evaluations in our study. This is expected because Tambur is designed for deadline-constrained streaming under bursty packet loss [55], whereas FlexFEC and ULPFEC are generic packet-level FEC mechanisms for the real-time transport protocol (RTP). Under bursty losses, multiple source packets within the same protection window can be lost together, making the repair packets insufficient to reconstruct all missing data. This issue is especially harmful to semantic-based telepresence: if the compact motion semantics of a frame cannot be recovered before the playback deadline, the renderer lacks complete semantic input, and playback stalls. Prior work also observes that Tambur outperforms WebRTC FEC for loss recovery in real-time 2D video streaming [15].

Video Smoothness. As shown in Figures 9(c) and 10(c), LITE consistently sustains over 30 FPS on average across both traces. The small variance in LITE’s FPS also indicates that its playback smoothness remains stable across time. In contrast, the baseline systems run below 20 FPS on average. Results under controlled loss rates shown in Figure 11(c) further demonstrate LITE’s advantage over the baselines. LITE achieves stable FPS under loss, whereas baseline systems degrade sharply as the loss rate increases. For example, they drop below 20 FPS at a 15% loss.

LITE achieves such a stable FPS because its recovery model reconstructs missing motion semantics within the playback deadline. For MagicStream, playback stalls occur when the applied FEC scheme fails to recover semantics, leading to FPS drops. For FarfetchFusion and MetaStream, FPS drops arise from two factors: (1) frames lost

during transmission cannot be recovered, and (2) the reduced sending rates imposed by congestion control under loss are insufficient to sustain the high-bandwidth demands of raw 3D content (§2.2). In addition to lower FPS, baseline systems show large variance. Such variability exacerbates playback unsmoothness and degrades user experience, as reflected in the subjective ratings presented later.

User Experience. Figures 9(d) and 10(d) present user ratings. LITE achieves an average score above 6.0 across both traces, surpassing baselines by 38% (6.5 vs. 4.7 for MA-T under the Hairpin trace) to 109% (6.2 vs. 3.0 for ME-U under the Akamai trace). Exit interviews indicate that users found LITE to provide satisfactory visual quality and a smooth viewing experience. In addition to poor visual quality, low and unstable frame rates were cited as another primary reason for the low ratings of baseline systems.

Bandwidth Consumption. LITE requires only 0.25 Mbps, because it transmits highly compact semantics and low-bitrate neural redundancy for audio recovery (§4.2). In contrast, MetaStream and FarfetchFusion transmit compressed raw 3D content, resulting in significantly higher bandwidth requirements. Moreover, FEC mechanisms introduce further overhead. For example, MetaStream with FlexFEC consumes 64.2 Mbps under the Hairpin trace, 253× higher than LITE, while FarfetchFusion with FlexFEC consumes 55.1 Mbps.

Audio Quality. Figure 15 presents user ratings of audio quality for LITE and various baselines on the Akamai trace. Results from the Hairpin trace exhibit similar trends. LITE achieves the highest user ratings (M: 6.5, SD: 0.2), outperforming LITE-L (M: 3.6, SD: 0.5) by 81%, Opus [61] (M: 4.2, SD: 0.4) by 55%, and Valin *et al.* [62] (M: 4.5, SD: 0.5) by 44%. This improvement is attributed to LITE’s neural redundancy, which effectively recovers audio under lossy networks. In contrast, LITE-L and Valin *et al.* rely only on previous audio semantics for prediction, leading to accumulated errors when consecutive frames are lost. Similarly, Opus with LBRR incorporates redundancy from only a single previous frame, making it ineffective in handling consecutive packet losses.

Performance against Relaxed-deadline Baselines. We further evaluate a relaxed-deadline setting to examine whether LITE remains effective when FEC-based baselines are given additional time for recovery. In this experiment, LITE keeps its default 100 ms playback deadline, while all baselines are evaluated under a relaxed 150 ms per-frame playback deadline. This setting is favorable to the baselines. In particular, Tambur can use a larger decoding-delay parameter: with around 80 ms spent on 3D processing and 30 FPS streaming (33 ms per frame), a 150 ms deadline allows Tambur to wait for up to two additional frames (*i.e.*, $\tau = 2$). Similarly, FlexFEC

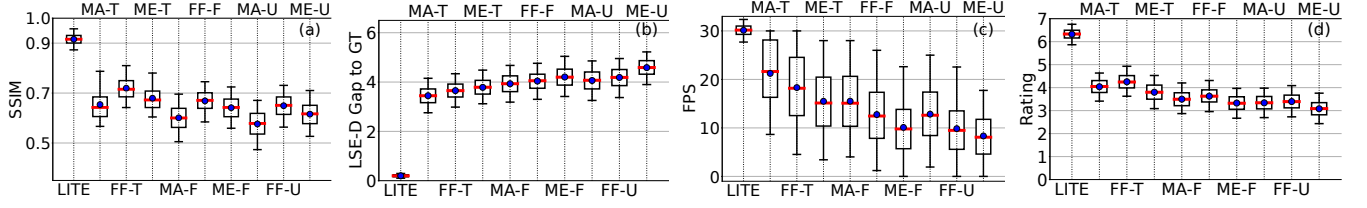


Figure 13: (a) SSIM, (b) LSE-D gap to ground truth, (c) FPS, and (d) user ratings on visual quality across LITE and the nine baseline systems under packet loss traces from Akamai [25]. LITE is evaluated under the 100 ms playback deadline, while other baseline systems are evaluated under the relaxed 150 ms playback deadline.

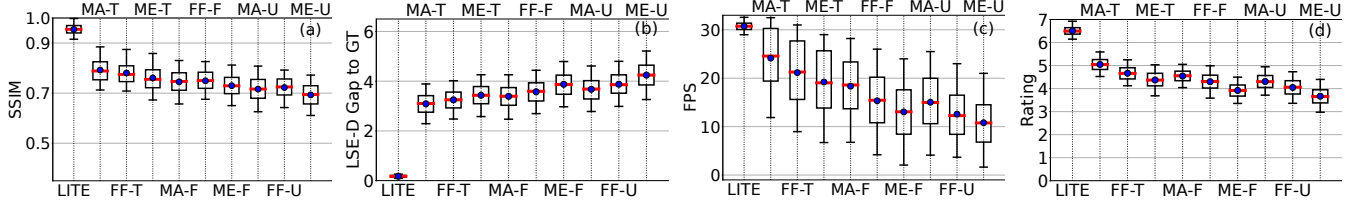


Figure 14: Relaxed-deadline evaluation under Hairpin packet-loss traces [47]. Metrics and setup are the same as in Figure 13.

and ULPFEC are also given more time to receive repair packets before playback. Moreover, this relaxed deadline setting reflects a possible future scenario in which 3D processing overhead is reduced, leaving more time for loss-recovery mechanisms.

Figures 13 and 14 present the results under the Akamai and Hairpin traces, respectively. We observe that, even under this favorable setting for the baselines, LITE consistently achieves the best performance. Under both traces, LITE maintains SSIM above 0.9, an LSE-D gap to ground truth near zero, over 30 FPS, and user ratings above 6.0. In contrast, even the best baseline remains below 0.8 SSIM, has an LSE-D gap above 3, runs below 25 FPS on average, and receives user ratings around 5.0. These results further demonstrate the advantage of LITE over existing loss-recovery mechanisms.

6.3 Ablation Study

In this section, we conduct an ablation study to quantify the contribution of each key component in LITE. We compare LITE with three variants: LITE-M/-A, which recovers motion semantics using only historical motion/audio semantics, as well as LITE-D, which directly concatenates audio and motion semantics as input to the recovery model (*i.e.*, without the modality-specific encoders and the fusion network). Starting from this section, we report results under the Akamai trace, which exhibits more severe loss than the Hairpin trace. Results under the Hairpin trace show the same trend.

As shown in Figure 16, using only motion or audio semantics results in only < 0.76 SSIM on average. In contrast, concatenating audio and motion semantics improves SSIM to ~ 0.8 on average, demonstrating the benefit of exploiting both modalities for motion recovery. Moreover, compared to LITE-D, LITE further increases SSIM to > 0.9 on average, highlighting the effectiveness of LITE’s design of modality-specific encoders and fusion network.

6.4 Parameter Sensitivity

We next evaluate LITE’s sensitivity to design parameters and analyze the trade-offs introduced by different configurations.

Multimodal Loss Recovery Model (§4.1). We analyze the impact of window size (*i.e.*, the number of previous frames used for loss

recovery) for motion and audio semantics. We evaluate motion windows of $\{1, 3, 5\}$ and audio windows of $\{4, 7, 10\}$. Figure 17 depicts the SSIM for these configurations. LITE, configured with a motion window of 3 and an audio window of 7, achieves the highest SSIM across all setups. This corresponds to a ~ 100 ms window for both semantics, indicating that approximately 100 ms of historical context carries the highest predictive value for the present frame.

Neural Redundancy (§4.2). We evaluate the impact of ℓ , the number of prior frames encoded into each neural redundancy. We test $\ell = \{4, 7, 11, 15\}$, corresponding to roughly 50–200 ms of historical data. Figure 18 reports user ratings of audio quality. The highest ratings occur at $\ell = 7$. User interviews reveal that $\ell = 4$ often produces audible gaps, since consecutive losses longer than four frames (≈ 53 ms) cannot be recovered. By contrast, $\ell = 7$ can tolerate up to seven consecutive frame losses (≈ 93 ms), yielding clearer and more continuous audio. However, increasing ℓ beyond 7 reduces user ratings. For $\ell = 11$ and $\ell = 15$, participants reported that the recovered audio sounded unnatural and poorly synchronized with the original speech, to the point that they preferred silence. This degradation arises from a fundamental property of audio communication rather than a limitation of LITE, as recovery beyond 100 ms becomes increasingly error-prone and perceptually inaccurate [62]. This also aligns with LITE’s 100 ms playback-deadline target: recovering content after this deadline provides little benefit for real-time telepresence, since frames rendered after their playback deadlines are considered lost [8, 15, 47].

We then evaluate the effectiveness of RVQ in reducing bandwidth consumption. Specifically, we test different codebook sizes (4, 8, 16, and 32), as well as a baseline without RVQ, where all raw prior audio semantics are transmitted as redundancy, consuming more bandwidth than RVQ-based schemes. Figure 19 shows that using a codebook size of 8 yields user ratings comparable to larger sizes (*e.g.*, 6.35 for LITE vs. 6.42 for no RVQ) while outperforming a size of 4 by 17.6% (6.35 vs. 5.40). Thus, we select a codebook size of 8 for LITE, reducing the bandwidth for audio redundancy to 0.006 Mbps, 24 $\times$ lower than the no-RVQ baseline.

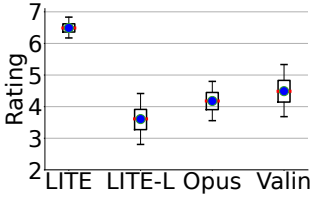


Figure 15: User ratings of audio quality among LITE, LITE-L (using LSTM to predict lost frames), Opus [61], and Valin *et al.* [62]

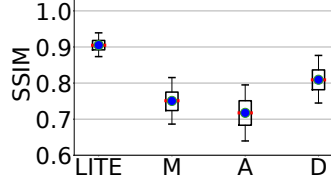


Figure 16: SSIM of LITE and its variants: M/A: using only historical motion/audio semantics; D: direct concatenation of audio and motion semantics.

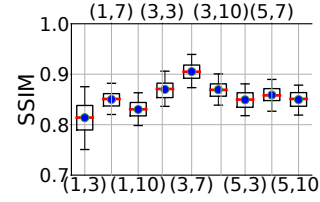


Figure 17: SSIM of multi-modal loss-recovery model for motion semantics in LITE with different ℓ (the number of prior frames encoded into each neural redundancy).

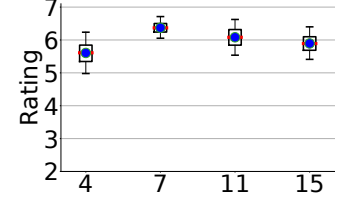


Figure 18: User ratings of audio quality of LITE with different ℓ (the number of prior frames encoded into each neural redundancy).

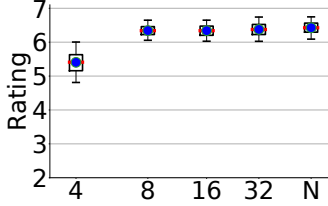


Figure 19: User ratings of audio quality across different codebook sizes of RVQ and without RVQ (N).

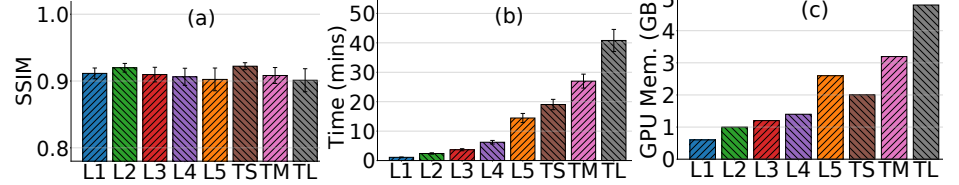


Figure 20: (a) SSIM, (b) training time (minutes), and (c) GPU memory (GB), when training the loss recovery model for motion semantics in LITE with LSTMs of one to five layers [26] (L1-L5) and Transformers [63] with small (TS), medium (TM), and large (TL) parameters.

6.5 Training Time, Inference Latency, and Model Generalizability

In this section, we evaluate the training time and inference latency of loss-recovery models in LITE. We further examine generalizability by testing whether, for returning users, LITE can fine-tune from an existing profile rather than training from scratch. We find that training the loss-recovery models for a new user takes only 1.8/1.6 minutes (SD: 0.2/0.2) on average for motion/audio semantics. Also, inference latency is only 2.7/2.4 ms (SD: 0.3/0.2) per frame on average. This fast inference time ensures that each lost frame is recovered before its deadline, maintaining a stable frame rate. The fast training and inference can be attributed to two factors: (1) the input to the loss-recovery models is highly compact; and (2) the model is LSTM-based, which is well-suited for real-time inference [67]. Note that all training is performed on the edge device. In practice, a service provider can instead perform training on a backend server equipped with more powerful GPUs, thereby further reducing training time.

We next evaluate the generalizability of LITE. When returning users start new telepresence sessions, LITE is able to fine-tune the trained models rather than training from scratch. This is because the recovery models capture user-specific characteristics, such as facial features and articulation patterns. Across different sessions, these factors remain highly consistent, requiring the model to adapt only to minor variations. To evaluate this, we ask one participant to complete five telepresence sessions, each with a different appearance (*e.g.*, different eyeglasses) and different spoken content during the profiling phase. We repeat each condition five times with a randomized order. Our results show that fine-tuning the model for the final session takes only 30 seconds on average for both motion and audio recovery. This reduction in training time highlights LITE's generalizability.

6.6 Impact of Model Architecture

We next evaluate how the backbone architecture of the loss-recovery model for motion semantics affects reconstruction quality and training cost. We consider two families of deep learning models commonly used to capture temporal dependencies: LSTM [26] and Transformer [63]. For LSTM, we vary the depth from one to five layers (L1–L5). For Transformer, we evaluate three model sizes following its design [63]: small (TS), medium (TM), and large (TL).

Figure 20 presents the SSIM, training time, and GPU memory consumption across model configurations. As shown in Figure 20(a), the two-layer LSTM (L2) and the small Transformer (TS) achieve the highest SSIM (~ 0.92). Given that L2 requires less training time (1.8 vs. 19 minutes) and GPU memory (1.0 vs. 2.0 GB), as shown in Figures 20(b) and (c), we adopt L2 as the backbone for motion-semantic recovery.

We also observe that larger models (L3–L5 and TM/TL) slightly reduce SSIM compared to L2/TS despite longer training time and higher GPU memory consumption. This may be due to overfitting: given that the recovery model operates on compact semantics, increasing model complexity may fit idiosyncrasies in the training data without improving generalization [73]. Also, as shown in Figure 20(a), all models achieve SSIM above 0.9, indicating that the chosen semantics and our multimodal recovery design provide informative inputs for loss recovery.

6.7 Performance under QUIC with CUBIC and Reno

We also evaluate whether LITE remains effective with different transport protocols and congestion-control algorithms. Figure 21 shows the results over QUIC using CUBIC and Reno for LITE, FarfetchFusion, and MetaStream. The latter two systems are configured with Tambur, which performs better than WebRTC-based FEC (§6.2).

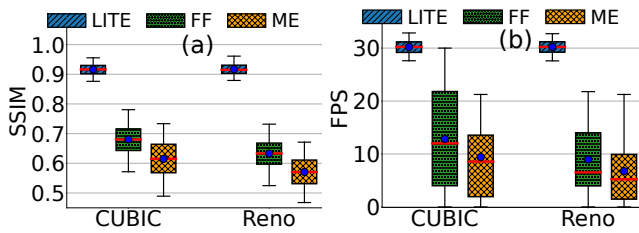


Figure 21: SSIM and FPS for LITE, FarfetchFusion (FF) with Tambur, and MetaStream (ME) with Tambur under Akamai's packet-loss traces when using QUIC with CUBIC or Reno.

In both cases, LITE consistently achieves an SSIM above 0.9 and an average FPS above 30. This robustness stems from LITE's low bandwidth requirement (0.25 Mbps, as evaluated in §6.2), which makes it insensitive to congestion-control dynamics. We also observe that Reno yields lower SSIM and FPS than CUBIC for FarfetchFusion and MetaStream. This is because Reno reacts more conservatively to packet loss and ramps up more slowly, resulting in lower sending rates than CUBIC (§2.2).

6.8 Microbenchmarking, Resource Utilization, and Scalability

We microbenchmark LITE on three edge devices: 1) a desktop with an RTX 4090 GPU, 2) a desktop with an RTX 3060 GPU, and 3) a Jetson AGX Orin embedded system. On these devices, LITE incurs average per-frame recovery latency of 2.7 ms, 6.2 ms, and 9.8 ms, respectively. Even on the Jetson embedded system, recovery remains fast enough to meet the playback deadline.

We next evaluate the resource usage of LITE on both the edge device and the mobile headset. The loss-recovery model in LITE utilizes 2.0 GB of GPU memory, < 8% CPU, and < 10% RAM on each tested edge device for two-way communication between two users. To assess scalability, we run multiple instances of the loss-recovery model concurrently on the same edge device. Our results show that the RTX 4090 and RTX 3060 devices, and the Jetson embedded system, can sustain 10, 5, and 3 concurrent users per side, respectively, without degrading LITE's performance.

To measure on-device battery usage, we fully charge a Microsoft HoloLens 2 headset and deploy it in a telepresence session for 1 hour. After the experiment, the battery level decreases to 72%, with average CPU and GPU utilization of 27% and 29%, respectively. Overall, we believe that LITE's resource utilization and energy consumption are acceptable.

7 Discussion

Recovering Full-body Immersive Content. LITE currently focuses on recovering motion semantics for the facial region, as detailed facial expressions are crucial for immersive telepresence [14, 37, 38]. A natural next step is to extend LITE to full-body motion semantics. This could be achieved by recovering parameters of the SMPL-X model [53]. A recent study [51] demonstrates that plausible full-body poses can be generated from audio alone, suggesting a potential link between speech and body motion. We are currently exploring methods to extract relevant audio features that can enable full-body motion semantics recovery.

Deployment. We have demonstrated that LITE provides an effective loss-recovery solution for immersive telepresence. Here, we outline key considerations for deploying LITE in practice. In real-world systems, it is common to simultaneously employ multiple loss-recovery techniques (e.g., retransmission and FEC) to protect data [47]. This is because the cost of data loss may outweigh the additional overhead these mechanisms incur. LITE is compatible with such hybrid deployments. For example, when combined with FEC, LITE can focus on recovering frames that FEC fails to correct. When retransmission is enabled, LITE's multimodal recovery model may benefit from a higher likelihood of receiving more recent ground-truth data, potentially improving reconstruction accuracy. Naturally, integrating FEC and retransmission with LITE incurs additional overhead. Determining the most effective configuration requires further empirical evaluation.

Semantic Hallucination. LITE mitigates semantic hallucination by training the motion-recovery model with ground-truth semantics captured during user profiling. Our LSE-D results further show that the recovered facial motion remains well aligned with speech under packet loss. Nevertheless, semantic hallucination can still occur because speech-to-face mapping is inherently one-to-many: the same acoustic signal may correspond to different facial expressions depending on context, personality, and intent [59]. In such cases, LITE may generate facial motion that is temporally plausible and lip-synchronized, but inconsistent with the speaker's intended expression. A potential direction is to incorporate other semantic constraints, such as text embeddings from online speech recognition [21] or sentiment and emotion embeddings inferred from speech or transcribed text [34, 58]. These signals could provide additional context about utterance meaning and emotional intent. However, online transcription, language understanding, and emotion inference introduce additional computation latency. Thus, designing mechanisms that effectively leverage these signals to reduce semantic hallucinations while preserving timely recovery remains an important research challenge.

8 Conclusion

In this paper, we presented the design, implementation, and evaluation of LITE, a first-of-its-kind loss-resilient immersive telepresence system that recovers motion and audio data over lossy networks. LITE utilizes a multimodal loss-recovery model for motion semantics that jointly leverages prior audio and motion semantics via dedicated encoders, followed by a fusion network. It further introduces a neural redundancy framework that encodes multiple audio semantics into succinct representations, enabling reconstruction of lost audio while introducing minimal bandwidth overhead. These innovations empower LITE to accurately and rapidly recover semantics in the presence of severe packet loss, enabling real-time streaming and high visual quality. Our thorough performance evaluations demonstrate that LITE significantly outperforms state-of-the-art solutions.

Acknowledgment

We thank the anonymous reviewers for their insightful feedback. This work was partially supported by the National Science Foundation under Grants CNS-2212296, CNS-2235049, CNS-2402992, and CNS-2437203.

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Appendix

A Details of Loss-recovery Model for Motion Semantics

In this section, we provide the implementation details of the loss-recovery model for motion semantics (§4.1). The modality-specific encoders and the fusion network are implemented as a two-layer LSTM [26], which yields the best performance among various model

architectures (§6.6). Each modality-specific encoder features a hidden dimension of 64, while the fusion network employs a hidden size of 256, followed by a linear projection to the target motion semantics θ . Training is performed using Adam [35] with a learning rate of 1×10^{-4} , a weight decay of 1×10^{-5} , and a batch size of 32. This specific configuration was selected after an extensive hyperparameter search, as it empirically yielded the optimal performance.

B Handling Color Semantic Loss

LITE does not currently recover color semantics (e.g., lighting characteristics [13]). This design choice follows our current deployment target: immersive telepresence is generally used in controlled lighting environments, where lighting conditions remain stable over time, as in commercial platforms such as Google Beam [5]. For future deployments in less controlled environments, a possible strategy is to transmit an additional front-view 2D video stream of the remote participant. This stream can serve as an auxiliary signal for detecting lighting changes and selecting or recovering color semantics. It can also be protected using existing 2D video loss-recovery methods [8, 15, 41, 47]. Specifically, by comparing the current front-view image with prior images using metrics such as SSIM, the receiver can determine whether prior color semantics can be safely reused. Alternatively, a multimodal model could be trained to recover lost color semantics using both prior color semantics and the front-view stream, similar to how LITE recovers motion semantics from prior motion and audio semantics (§4.1). This strategy introduces a trade-off, since transmitting a front-view video stream can incur additional bandwidth overhead, for example, over 5 Mbps [32]. Therefore, it should be enabled only when the residual bandwidth after congestion-control adaptation can still support the additional stream.

C Analysis of Hairpin and Tooth

In this section, we analyze Hairpin [47] and Tooth [8], two loss-recovery methods originally designed for cloud gaming, to explain why they are infeasible in our setting and therefore excluded from the performance evaluation (§6.1).

Hairpin dynamically adjusts the FEC redundancy rate in conjunction with retransmission. Specifically, it applies a small redundancy rate during the first transmission attempt, and if a packet is lost, it retransmits with a higher redundancy rate. Since retransmission is infeasible in our case (§6.1), Hairpin reduces to a standard FEC scheme, which performs worse than Tambur (§6.1), the FEC-based solution adopted in our evaluation.

Tooth dynamically assigns the FEC redundancy rate based on the number of packets required to transmit each frame. This mechanism is feasible for 2D video streaming, where modern codecs such as H.264 [3] use inter-frame compression, resulting in variable frame sizes. In contrast, this approach is not applicable to semantic communication, where each frame is compact and always transmitted as a single packet. Thus, in our case, Tooth also reduces to a standard FEC scheme.